Final Project - Submission 1

2026-10-17

```
library(ggplot2)

## import data
covarData <- read.csv("/Users/laurentuzik/Desktop/Dartmouth/FALL/HSE 103 - Foundations of Data Science/I

expData <- read.csv("/Users/laurentuzik/Desktop/Dartmouth/FALL/HSE 103 - Foundations of Data Science/Fi

namesData <- read.csv("/Users/laurentuzik/Desktop/Dartmouth/FALL/HSE 103 - Foundations of Data Science/I

#head(covarData)
#head(expData) ## need to make geoID into column and preseve X
#head(namesData)
```

```
## need to switch X (ProbeID) and geoID in expression data
## want to save the X column data
X <- expData$X
## first - need to make ProbeID/X the row names
## want to keep X attached to its corresponding data
rownames(expData) <- expData$X
#head(expData)
## then need to delete the X column because it is now the row names
expData <- expData[ , -1]
#head(expData)
## then transpose the data - so the rows become columns/columns become rows
expData_t <- as.data.frame(t(expData))
#head(expData_t)
## lost the X so i can restore
#colnames(expData_t) <- X
#head(expData_t)

## change the probeID to gene name in the merged sheet
# ?match()
names(expData_t) <- namesData$GENE[match(names(expData_t), namesData$PROBEID)]
#head(expData_t)

## need to add a geoID label on the first cell of the sheet
expData_t$geoID <- rownames(expData_t)
#head(expData_t)

## can then merge data on geoID variable
merged <- merge(covarData, expData_t, by = "geoID")
```

```
## All submitted plots must be made using ggplot2 and must be of publication quality
## Identify one gene, one continuous covariate, and two categorical covariates of interest.
## At least one categorical variable should be derived from a continuous variable
```

```r
## gene - NPPB
## one continuous covariate - mat_age_years
## two categorical covariates, one must be derived -
## preg_preclampsia, mat_vitD_group

# class(merged$mat_vitD) - its chr - will need to change to numeric so i can group
# head(merged$mat_vitD) - contains a string before the numbers - will need to remove
## substitute everything up to : with "" empty space in the mat_vitD variable from merged
merged$mat_vitD <- as.numeric(sub(".*:", "", merged$mat_vitD))

## then subset into groups for the derived categorical variable from a continuous
## low = <20; high = >=20
merged$mat_vitD_group <- ifelse(merged$mat_vitD <20, "Low", "High")

#class(merged$mat_age_yrs) # is chr - need to change to numeric
## can use same format as vitamin d
merged$mat_age_yrs <- as.numeric(sub(".*:", "", merged$mat_age_yrs))
#head(merged$mat_age_yrs)

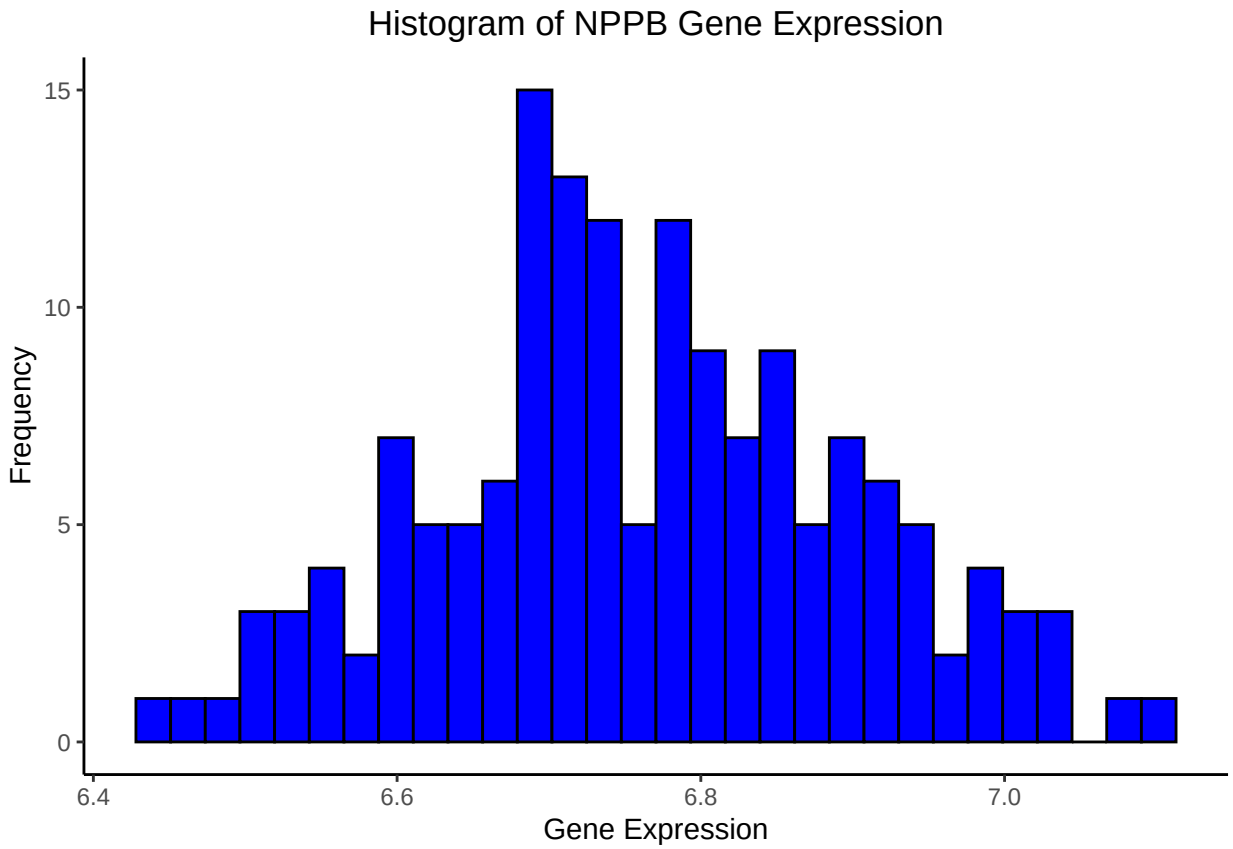
# head(merged$preg_preeclampsia) - need to change to normal and preclampsia only
## use sub
merged$preg_preeclampsia <- sub(".*:", "", merged$preg_preeclampsia)
#head(merged$preg_preeclampsia)
```

```r
## define a nice theme for my plots

my_theme <- theme(
        # Remove all the extra borders and grid lines
        panel.border = element_blank(), panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        # Define  axis
        axis.line = element_line(colour = "black", linewidth = rel(1)),
        # Set plot background
        plot.background = element_rect(fill = "white"),
        panel.background = element_blank(),
        legend.key = element_rect(fill = 'white'),
        # Move legend
        legend.position = 'top')
```

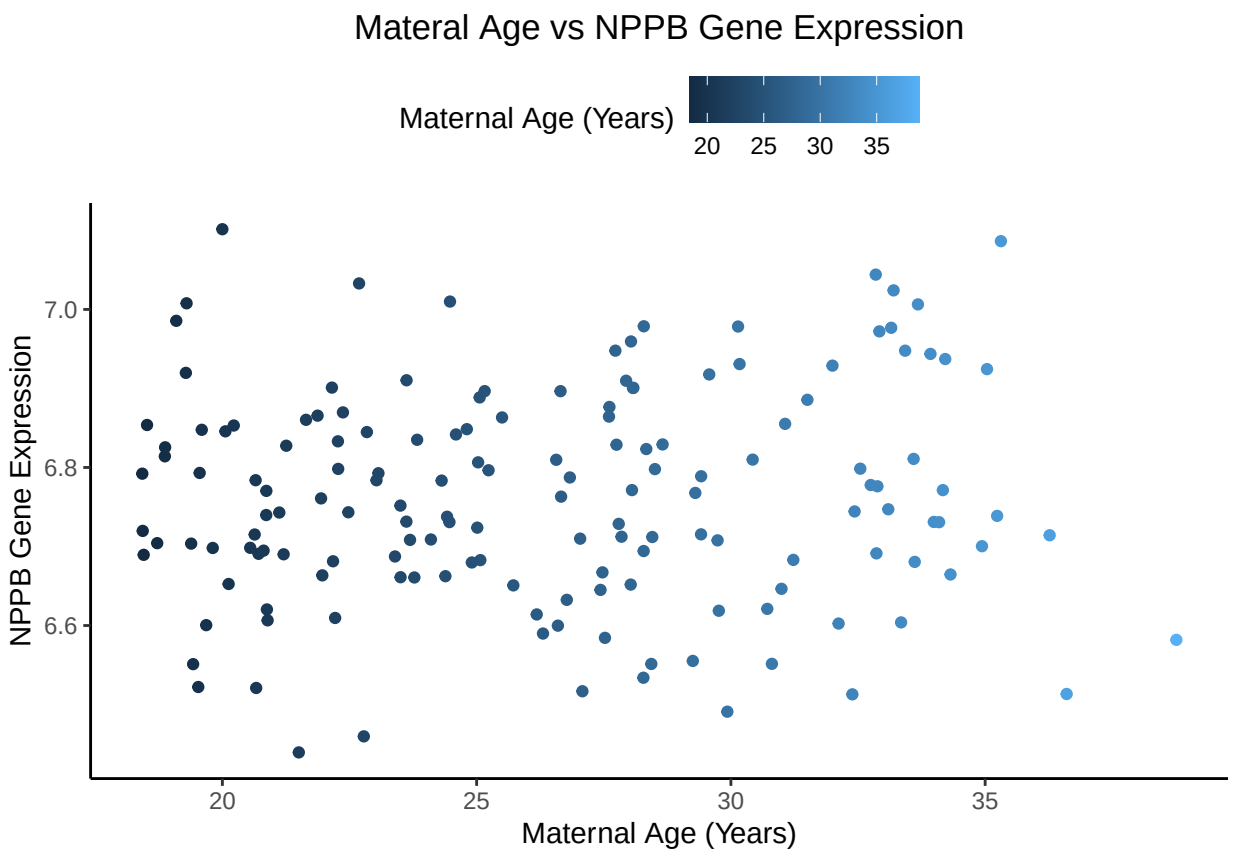
```r
## 1 Histogram of Gene Expression
# library(ggplot2)
ggplot(merged, aes(x = NPPB))+
  # fill blue with black outline
  geom_histogram(fill = "blue", color = "black") +
  labs(title = "Histogram of NPPB Gene Expression", x = "Gene Expression", y = "Frequency")+
  my_theme +
  # want a centered title
  theme(plot.title = element_text(hjust = 0.5))
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



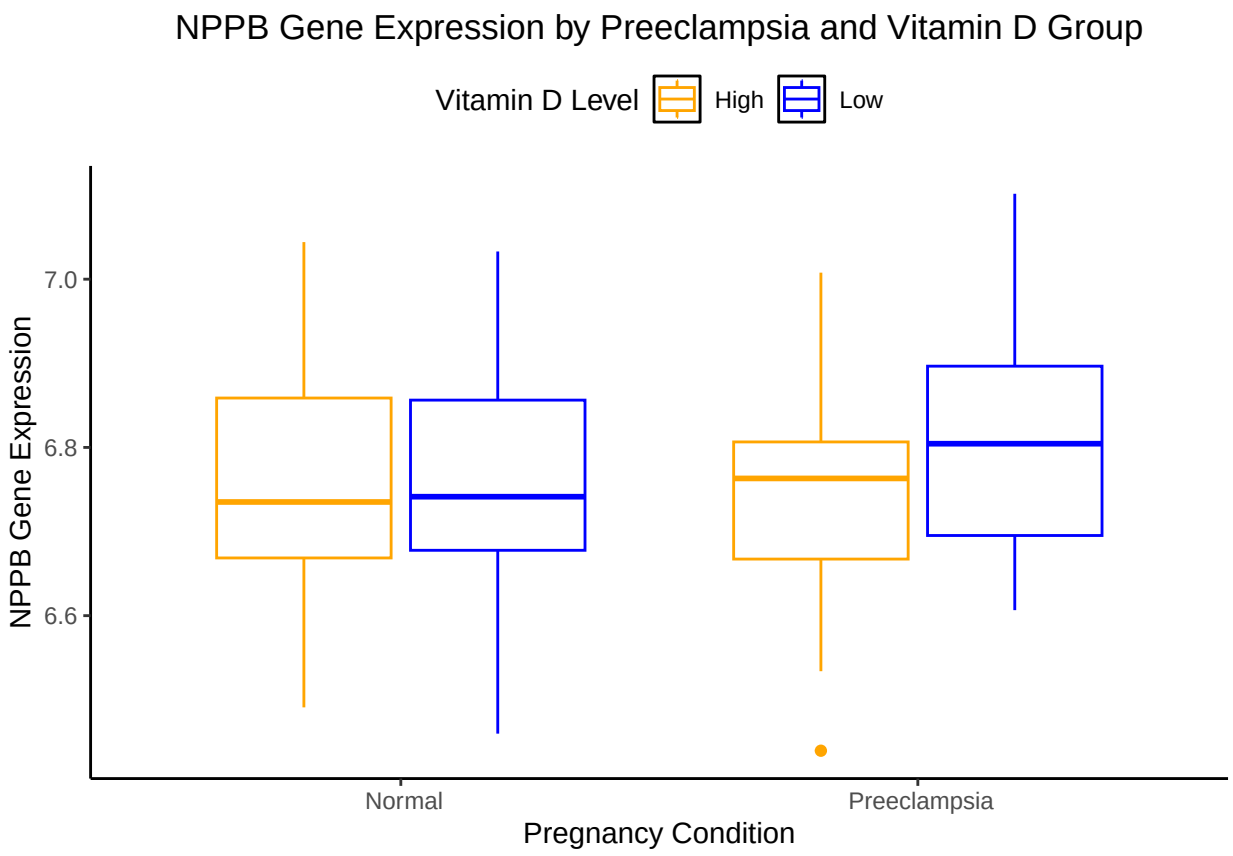

```
## 1 Scatterplot of Covariate vs. Gene Expression
# class(merged$mat_age_yrs) - is chr - go back and change to numeric

ggplot(merged, aes(x = mat_age_yrs, y = NPPB, color = mat_age_yrs))+
  geom_point()+
  labs(title = "Materal Age vs NPPB Gene Expression", x = "Maternal Age (Years)",
       y = "NPPB Gene Expression",
       color = "Maternal Age (Years)") +
  my_theme +
  theme(plot.title = element_text(hjust = 0.5))
```




```
## 1 Boxplot of Gene Expression Separated by Both Categorical Variables

ggplot(merged,aes(x = preg_preeclampsia, y = NPPB, color = mat_vitD_group)) +
  geom_boxplot()+
  labs(title = "NPPB Gene Expression by Preeclampsia and Vitamin D Group",
       x = "Pregnancy Condition", y = "NPPB Gene Expression", color = "Vitamin D Level") +
  scale_color_manual(values = c("orange", "blue")) +
  my_theme +
  theme(plot.title = element_text(hjust = 0.5))
```


NPPB Gene Expression by Preeclampsia and Vitamin D Group
Vitamin D Level
High
Low
NPPB Gene Expression
7.0
6.8
6.6
Normal
Preeclampsia
Pregnancy Condition